The spacetime diagrams shown below describe the implosion of a spherical star to form a spherical black hole; compare with Figure 6.7. The dotted curves are *outgoing light rays;* in other words, they are the world lines (trajectories through spacetime) of photons—the fastest signals that can be sent radially outward, toward the distant Universe. For optimal escape, the photons are idealized as not being absorbed or scattered at all by the star's matter.

The *apparent horizon* (left diagram) is the outermost location where outgoing light rays, trying to escape the hole, get pulled inward toward the singularity (for example, the outgoing rays QQ′ and *RR′).* The apparent horizon is created suddenly, full-sized, at *E,* where the star's surface shrinks through the critical circumference. The *absolute horizon* (right diagram) is the boundary between events that *can* send signals to the distant Universe (for example, events *P* and *S* which send signals along the light rays *pp′* and *SS′)* and events that *cannot* send signals to the distant Universe (for example, Q and *R).* The absolute horizon is created at the star's center, at the event labeled C, well before the star's surface shrinks through the critical circumference. The absolute horizon is just a point when created, but it then expands smoothly, like a balloon being blown up, and emerges through the star's surface precisely when the surface shrinks through the critical circumference (the circle labeled *E).* It then stops expanding, and thereafter coincides with the suddenly created apparent horizon.

